

Venus Flashloans Fix Review Report

Please Note

1. The analysis of the Severity is purely based on the smart contracts mentioned in the Audit Scope and does not include any other potential contracts deployed by the Owner. No applications or operations were reviewed for Severity. No product code has been reviewed.
2. Due to the time limit, the audit team did not do much in-depth research on the business logic of the project. It is more about discovering issues in the smart contracts themselves.

Audit Period: 2025/10/15 - 2025/10/16 (YYYY/MM/DD)

Smart contracts list:

No.	Contract Name	Link	Verdict	Details
1	FacetBase.sol	https://github.com/VenusProtocol/venus-periphery/blob/39a501a14bf134c743bd9ad3e538f62b34bb2a63/contracts/PositionSwapper/ISwapHelper.sol	Green	
2	FlashLoanFacet.sol	https://github.com/VenusProtocol/venus-periphery/blob/39a501a14bf134c743bd9ad3e538f62b34bb2a63/contracts/PositionSwapper/PositionSwapper.sol	Low	[L01] [I01] [I02] [I03]
3	SetterFacet.sol	https://github.com/VenusProtocol/venus-periphery/blob/39a501a14bf134c743bd9ad3e538f62b34bb2a63/contracts/PositionSwapper/WBNBSwapHelper.sol	Green	
4	IFlashLoanFacet.sol	https://github.com/VenusProtocol/venus-protocol/blob/f0ab93f24bdd2659827856c3c778b07f3f83d0b0/contracts/Comptroller/Diamond/facets/FacetBase.sol	Green	
5	ISetterFacet.sol	https://github.com/VenusProtocol/venus-protocol/blob/f0ab93f24bdd2659827856c3c778b07f3f83d0b0/contracts/Comptroller/Diamond/facets/MarketFacet.sol	Green	
6	Diamond.sol	https://github.com/VenusProtocol/venus-protocol/blob/f0ab93f24bdd2659827856c3c778b07f3f83d0b0/contracts/Comptroller/Diamond/facets/PolicyFacet.sol	Low	[L02]
7	ComptrollerInterface.sol	https://github.com/VenusProtocol/venus-protocol/blob/f0ab93f24bdd2659827856c3c77	Green	

		8b07f3f83d0b0/contracts/Comptroller/Diamond/facets/SetterFacet.sol		
8	ComptrollerStorage.sol	https://github.com/VenusProtocol/venus-protocol/blob/f0ab93f24bdd2659827856c3c778b07f3f83d0b0/contracts/Comptroller/Diamond/Diamond.sol	Green	
9	IProtocolShareReserve.sol	https://github.com/VenusProtocol/venus-protocol/blob/f0ab93f24bdd2659827856c3c778b07f3f83d0b0/contracts/Comptroller/Diamond/interfaces/IMarketFacet.sol	Green	
10	IFlashLoanReceiver.sol	https://github.com/VenusProtocol/venus-protocol/blob/f0ab93f24bdd2659827856c3c778b07f3f83d0b0/contracts/Comptroller/Diamond/interfaces/IPolicyFacet.sol	Green	
11	VBep20Delegator.sol	https://github.com/VenusProtocol/venus-protocol/blob/f0ab93f24bdd2659827856c3c778b07f3f83d0b0/contracts/Comptroller/Diamond/interfaces/ISetterFacet.sol	Green	
12	VToken.sol	https://github.com/VenusProtocol/venus-protocol/blob/f0ab93f24bdd2659827856c3c778b07f3f83d0b0/contracts/Comptroller/ComptrollerInterface.sol	High	[H01] [I01]
13	VTokenInterfaces.sol	https://github.com/VenusProtocol/venus-protocol/blob/f0ab93f24bdd2659827856c3c778b07f3f83d0b0/contracts/Comptroller/ComptrollerStorage.sol	Green	

Internal Audit

Issue No.	Severity	Status
[H01]	High	Fixed
[L01]	Low	Fixed
[I01]	Informational	Acknowledged
[I02]	Informational	Acknowledged
[I03]	Informational	Acknowledged
[I04]	Informational	Fixed

[H01] accrueInterest missing flashLoanAmount accounting

Contract VToken.sol

Severity Level High

Commit <https://github.com/VenusProtocol/venus-protocol/pull/638/commits/194b387296bddae3dfcfe1b8cca1ebc46d98ee6e>

Status Fixed, introduced the new _getCashPriorWithFlashLoan function that accounts for the current flashLoanAmount. This is now used in the following functions to ensure accuracy of current underlying asset balance

- accrueInterest()
- supplyRatePerBlock()
- borrowRatePerBlock()
- exchangeRateStoredInternal() → Now utilizes _getCashPriorWithFlashLoan to compute totalCash but the core logic is unchanged

[L01] Incorrect order of input validations leads to unhandled panic

Contract FlashloanFacet.sol

Severity Level Low

Commit <https://github.com/VenusProtocol/venus-protocol/commit/c9f1d56579ade5c5d13254a494b7ad7eb935ad51>

Status Fixed, now checks that the array lengths of vTokens and underlyingAmounts is equal before the flashloan enabled and 0 underlyingAmount checks

[I01] Interest not accrued to account for protocol fee gap

Contract FlashloanFacet.sol

Severity Level **Informational**

Commit -

Status Acknowledged, all market actions will ensure interest is up to date even if there is a protocol fee gap. This is acceptable and will only affect the current return value of the interest.

[I02] doTransferOut do not take into account FOT token characteristics

Contract FlashLoanFacet & VToken.sol

Severity Level **Informational**

Commit -

Status Acknowledged, Venus has accepted that this is an acceptable design decision where flashloan fees are charged on the full flashloaned amount even if the underlying asset has a transfer tax, given this amount would be a minor value. Additionally, Venus has indicated that they do not plan to support fee-on-transfer tokens.

[I03] Whitelisted flashloan contracts should have additional access control protections

Contract FlashloanFacet.sol

Severity Level **Informational**

Commit -

Status Acknowledged, Venus has accepted that this is an acceptable design decision where flashloan fees are charged on the full flashloaned amount even if the underlying asset has a transfer tax. Additionally, Venus has indicated that they do not plan to support fee-on-transfer tokens.

[I04] Missing documentation for the FlashLoanAlreadyActive error

Contract FlashloanFacet.sol

Severity Level **Informational**

Commit	https://github.com/VenusProtocol/venus-protocol/pull/638/commits/9f05cca562700f19152d88c4541bd58a15675ccf https://github.com/VenusProtocol/venus-protocol/pull/638/commits/4bb366a1c43f5b0af0ea5aa1fb1fb2c078b91f39
Status	Fixed, now includes documentation for both the NotAnApprovedDelegate and FlashLoanAlreadyActive error.

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Issue No.	Severity	Status
[VLW-08]	Centralization	Acknowledged
[VLW-09]	Major	Fixed
[VLW-11]	Medium	Fixed
[VLW-12]	Minor	Fixed
[VLW-13]	Minor	Fixed
[VLW-14]	Minor	Fixed
[VLW-05]	Minor	Fixed
[VLW-07]	Informational	Fixed
[VLW-10]	Informational	Acknowledged
[VLW-15]	Informational	Fixed
[VLW-16]	Informational	Acknowledged
[VLW-17]	Informational	Fixed
[VLW-01]	Optimization	Fixed
[VLW-02]	Optimization	Fixed
[VLW-04]	Optimization	Fixed
[VLW-06]	Optimization	Fixed

[VLW-08] Centralization Related Risks

Contract VToken.sol

Severity Level Centralization

Commit -

Status Acknowledged, Venus has sufficient Timelock addresses in place to control important role base access control.

[VLW-09] Manipulated Utilization Rate During Flashloan Can Cause Incorrect Interest Accrual

Contract VToken.sol

Severity Level **Major**

Commit <https://github.com/VenusProtocol/venus-protocol/pull/638/commits/194b387296bddae3dfcfe1b8cca1ebc46d98ee6e>

Status Fixed, see internal audit [H01]

[VLW-11] Missing Input Validation And Checks

Contract VToken.sol

Severity Level **Medium**

Commit <https://github.com/VenusProtocol/venus-protocol/commit/9821c492379b0b2b5f4f93b09a3dccc7914ad0d>

Status Fixed, now ensures that

- flashLoanFeeMantissa_ is within 100% represented by 1e18
- flashLoanProtocolShare_ is within 100% represented by 1e18
- amount to flashloan is less than or equal to available underlying asset balance

[VLW-12] Incorrect Supply And Borrow Rates Returned During Flashloan

Contract VToken.sol

Severity Level **Minor**

Commit <https://github.com/VenusProtocol/venus-protocol/pull/638/commits/194b387296bddae3dfcfe1b8cca1ebc46d98ee6e>

Status Fixed, see internal audit [H01]

[VLW-13] Flashloan Callback Does Not Include The Original msg.sender

Contract VToken.sol

Severity Level **Minor**

Commit <https://github.com/VenusProtocol/venus-protocol/commit/757e323c565bac753e7250662d562170e06d67de>

Status	Fixed, now includes a new initiator parameter within the executeOperation function.
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[VLW-14] Missing NonReentrant Modifier	
Contract	VToken.sol
Severity Level	Minor
Commit	https://github.com/VenusProtocol/venus-protocol/commit/869afd88210f03720a98fceb940c6e508cb727b1
Status	Fixed, now includes the nonReentrant modifier in the renamed flashLoanDebtPosition. Note that reentrancy will have no effects on repayments originally given the CEI pattern is correctly followed, but addition of the modifier is acceptable for safeguards

[VLW-05] Discussion On Updates To VBep20Delegator	
Contract	VBep20Delegator.sol
Severity Level	Minor
Commit	https://github.com/VenusProtocol/venus-protocol/commit/49382e4057c0f52199d37557cfcf5ff0ac8efdf0
Status	Fixed, borrowDebtPosition was renamed to flashLoanDebtPosition and removed from the VTokenInterface

[VLW-07] Discussion On Entities That Will Be Whitelisted For Flashloans	
Contract	FlashLoanFacet.sol
Severity Level	Informational
Commit	https://github.com/VenusProtocol/venus-protocol/commit/88872ebfdc8c33c41f56f474c2648bf4c0feb4dd
Status	Fixed, only vetted addresses will be whitelisted and allowed for flashloans. Additionally, the new transient reentrancy guard is included in the executeFlashloan function, disallowing any reentrancy back into the FlashloanFacet contract. However, note that since Venus operates on a diamond proxy pattern, in the future, if any new Facets are added with the nonReentrant modifier, a call to executeFlashloan will block calls into any specific facet logic that utilizes the same modifier, since a diamond proxy uses a single proxy contract that delegates calls to multiple facet implementations and share the same proxy storage.

[VLW-10] Minting During Flashloan Will Capture Some Of Suppliers Flashloan Fees

Contract VToken.sol

Severity Level Informational

Commit -

Status Acknowledged, accepted behavior that the user flashloaning gain some of the flash loan fees donated as supplier fees. This is acceptable, since the user minting new vTokens must still have returned the flashloaned underlying assets or risk a borrowed position, so the small amount of fee extracted that he paid himself is not sufficient to risk an interest payment for a borrowed debt position.

[VLW-15] Missing/Incomplete NatSpec Comments

Contract FlashloanFacet & VToken.sol

Severity Level Informational

Commit <https://github.com/VenusProtocol/venus-protocol/commit/9f05cca562700f19152d88c4541bd58a15675ccf>

Status Fixed, included natspec comments for all mentioned functions

[VLW-16] Integrators Relying On getCash() Should Ensure Compatibility With Flashloan Implementation

Contract VToken.sol

Severity Level Informational

Commit <https://github.com/VenusProtocol/venus-protocol-documentation/pull/309>

Status Acknowledged, WIP on including documentation for potential/existing external integrators that calls getCash()

[VLW-17] Typos/Inconsistencies

Contract VToken.sol

Severity Level Informational

Commit <https://github.com/VenusProtocol/venus-protocol/commit/9f05cca562700f19152d88c4541bd58a15675ccf>

Status Fixed, all typos/inconsistencies considered and applied where appropriate.

[VLW-01] Array Length Is Not Cached

Contract VToken.sol

Severity Level Optimization

Commit <https://github.com/VenusProtocol/venus-protocol/commit/9f05cca562700f19152d88c4541bd58a15675ccf>

Status Fixed as recommended

[VLW-02] Unused Struct Elements

Contract FlashloanFacet.sol

Severity Level Optimization

Commit <https://github.com/VenusProtocol/venus-protocol/commit/9f05cca562700f19152d88c4541bd58a15675ccf>

Status Fixed as recommended

[VLW-04] Unused Error

Contract ErrorReporter.sol

Severity Level Optimization

Commit <https://github.com/VenusProtocol/venus-protocol/commit/f05bf542e79b11d6cff7d7ce426a1dfefbcb6dc>

Status Fixed as recommended

[VLW-06] Unnecessary External Call For Flash Loan Amount

Contract VToken.sol

Severity Level Optimization

Commit <https://github.com/VenusProtocol/venus-protocol/commit/1c9dbd1c08a902ae0c4a1006a0ee96e164f1c025>

Status Fixed as recommended

Note that the only way for flashLoanAmount to differ from underlyingAmounts is for

the underlying asset token flashloaned to have fee-on-transfer or custom transfer logic.
